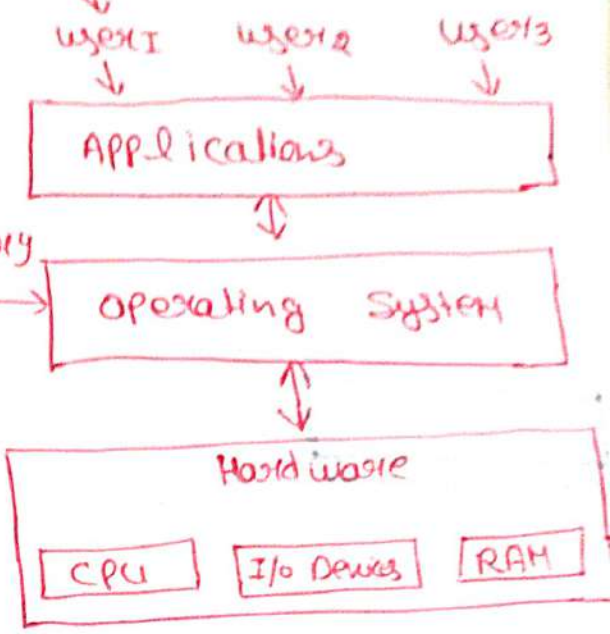
V-1 Operating System

Operating System :-

- (OS) as an interface between user and hardware. It is responsible for the execution of all the processes Resource Allocation, [CPU Management] intermediary File Management and Many other tasks.
- Examples of operating System are windows, Linux, MacOS, Android, ios etc.



OS used :-

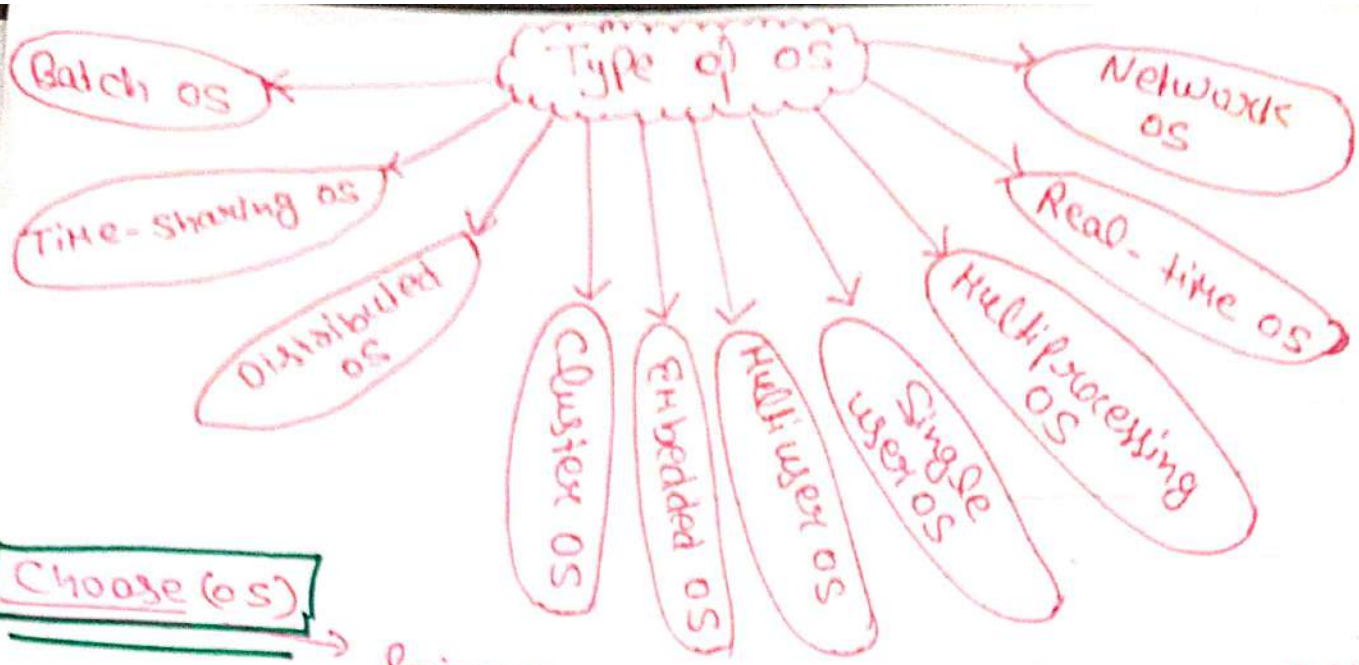
1) As a platform for Application program.
2) Managing (I/P) / (O/P) unit.
3) Multitasking
4) Manages Memory and files.
5) Provide Security.

Function of OS :-

1) Resource Management.
2) Process Man.....
3) Memory Man.....
4) Device Man.....
5) File Man.....
6) Time - Sharing.
7) Networking
8) Job Accounting.
9) Security
10) user interface
11) Backup and Recov
12) Performance Moni

objectives of OS :-

1) user friendly.
2) Easy Access.
3) Management of Resources.
4) Controls and Monitoring.

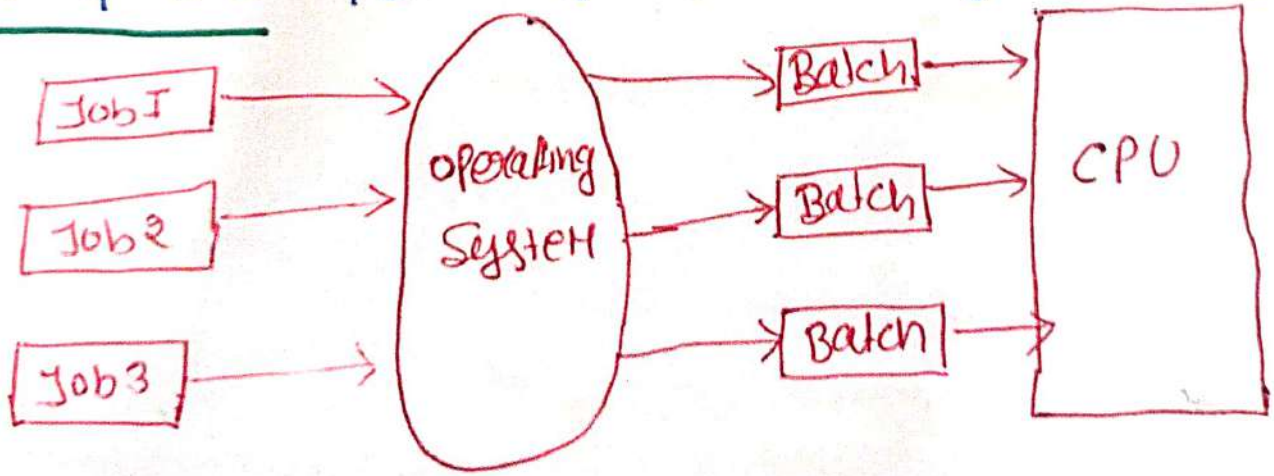


Choose (OS) → Price factor, Accessibility factor, Compatibility factors, Security factor.

examples of OS:-

Batch OS :- (Batch of Job)

- → Batch Processing was very popular in the 1970s.
- → users using batch (OS) do not interact directly with the computer.
- → Each user prepares their Job using an offline device like a punch card and submitting it to the computer operator.
- → there is an operator which takes similar Jobs having the same requirements and groups them into batches.
- Batch (OS) is designed to manage and execute a large number of Jobs efficiently by processing them in groups.



Advantages of Batch (OS) :-

- → Multiple users can share the batch system.
- → The idle time for the batch system is very less.
- → It is easy to manage large work repeatedly in batch system.

DS Advantages :-

- → CPU is not used effeciently
- → the other jobs will have to wait for an unkown time if any job fails.
- → In batch (OS), average response (T) increases as all process one by one.

Eg :- Payroll systems, Bank statements etc.

2. Multi-Programming (OS) :-

→ Multiprogramming is a technique to execute number of programs simultaneously by a single processor.

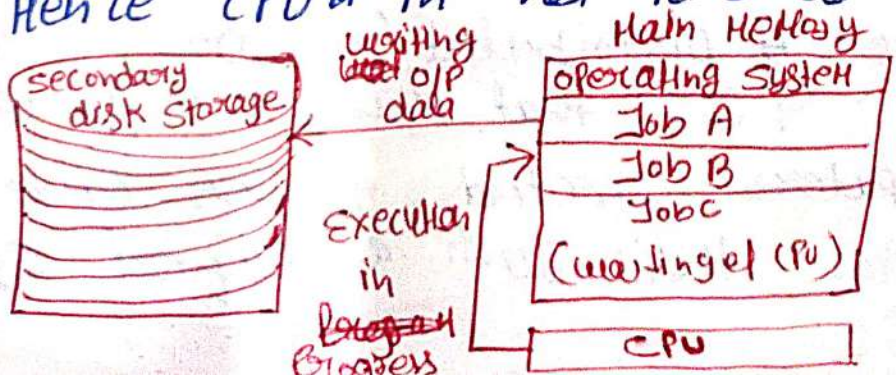
→ The objective of a Multiprogramming (OS) is to increase the system utilization efficiency.

→ In Multiprogramming number of process reside in main memory at a time.

→ The OS picks and begins to executes one of the jobs in the main memory.

→ if any I/O wait happened in a process, the CPU switch from that job to another job.

→ Hence CPU in in not idle at any time.



Advantages of Multi-programming OS :-

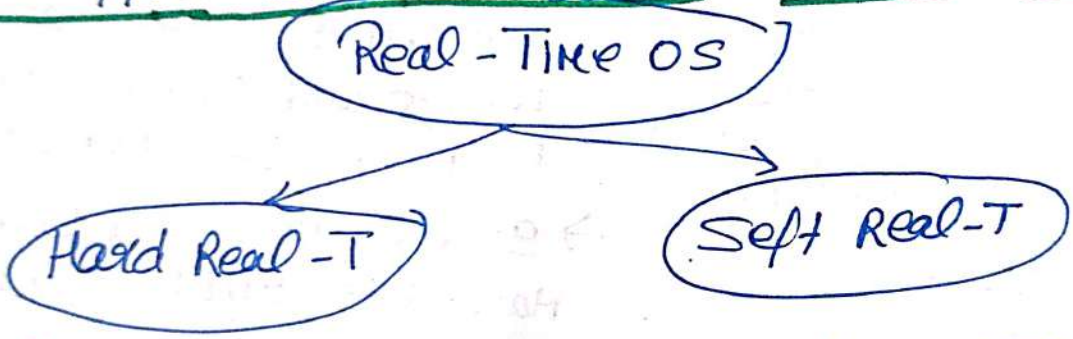
- CPU is better utilized and overall performance of the system improves.
- It helps in ~~reduce~~ reducing the response time.

Disadvantages :-

- It is highly complicated and sophisticated.
- The CPU scheduling is required.
- The harder task is to handle all processes and task.

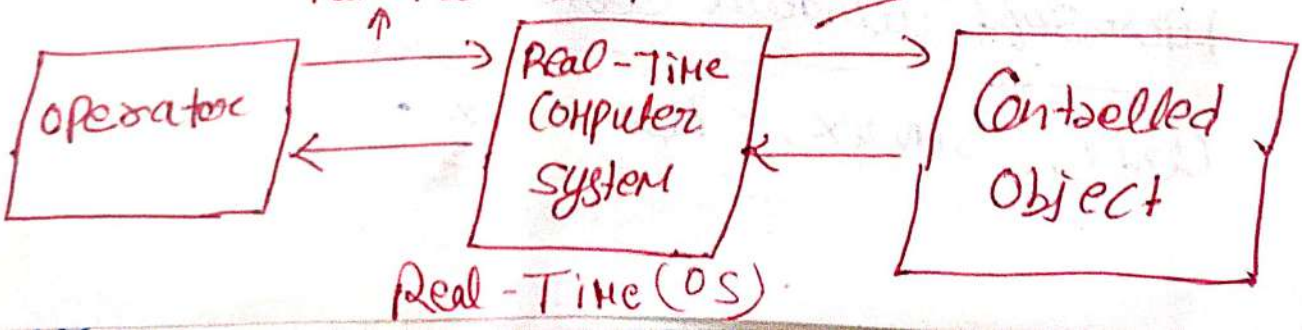
Real-Time (OS) :-

- these type of (OS) sevel real-time system. the (T) interval required to process and respond to inputs is very small. this (T) interval is called response time. ~~Real (T) System~~.
- Real-time (OS) :- are used when there are time requirements that are very strict missile system, air traffic control system, robots, etc.



Hard Real(T) :- (—) are meant for application where time constraints are very strict and even the shortest possible delay is not acceptable.

Soft Real(T) :- These OS are for application where constraint is less strict.



⇒ Embedded Systems :-

→ Arrangement.

→ Way of Working.

Something that is attached to another Thing.

→ Independent System / Part of a large System.

Components

- Hardware
- Software
- RTOS

Embedded OS :- Specialized OS designed to perform a specific task. for a device that is not a computer.

eg :- Embedded linux & Android OS.

Features :-

1) Small Processing Power.
2) Limited storage Capacity.
3) Fast & Limited functionalities.
4) Portable, energy Effecient.

Application :-

1) Smart Tv, Smart washing Machine, Smart AC
2) Traffic Lights, Mobile (PH). ATM Machine.
3) Air traffic Control, Digital Cameras.

Multiprogramming	Multi-tasking
• It includes the single CPU to execute the program.	• It uses multiple tasks for the task allocation within the same CPU, allowing it to manag it to manage and execute several operations concurrently.
✓ Concept of Context Switching is used.	• Concept of Context Switching & Time Sharing is used.
• Multi-programming increases CPU utilization by organizing Jobs.	• In Multi-tasking also increases CPU utilization, it also increases ~~responsive~~ responce time. — ~~less~~.
• The idea is to reduce the CPU idle time for as long as possible.	• the idea is to further extend the CPU utilization concept by increasing responsiveness time sharing.
• It used Job scheduling Algo so that more than one program can run at the same time.	• Time sharing mechanism is used so that multiple tasks can run at the same T
• It community edition. personalized shopping experiences is not created.	• Promotions, personalized shopping experiences can be displayed in enterprise edition products.
• Executions of process takes more time.	• Execution of process takes ~~more~~ less time.
• Simple to implement.	• Complex to implement.
• objective is to	

Process States

Process :- • Program under execution is known as Process.

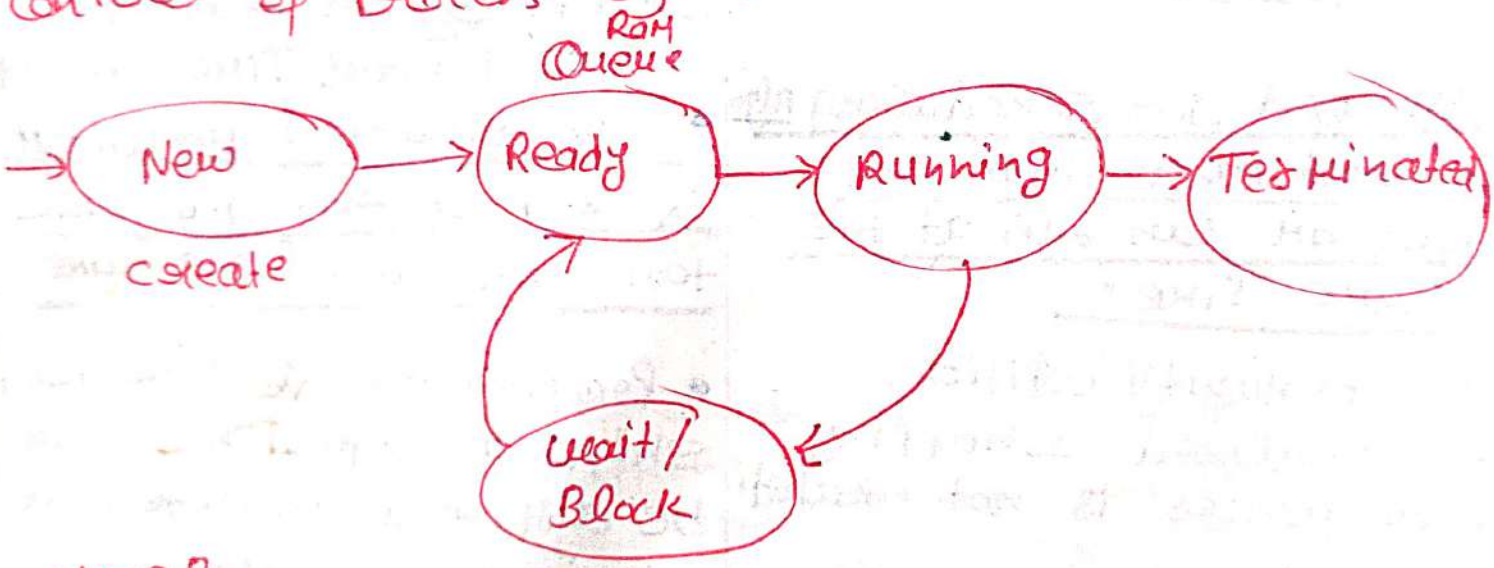
• Program is passive entity stored on disk (executable file). Process is active.

→ Program becomes Process when executable file loaded into Memory.

Process states :-

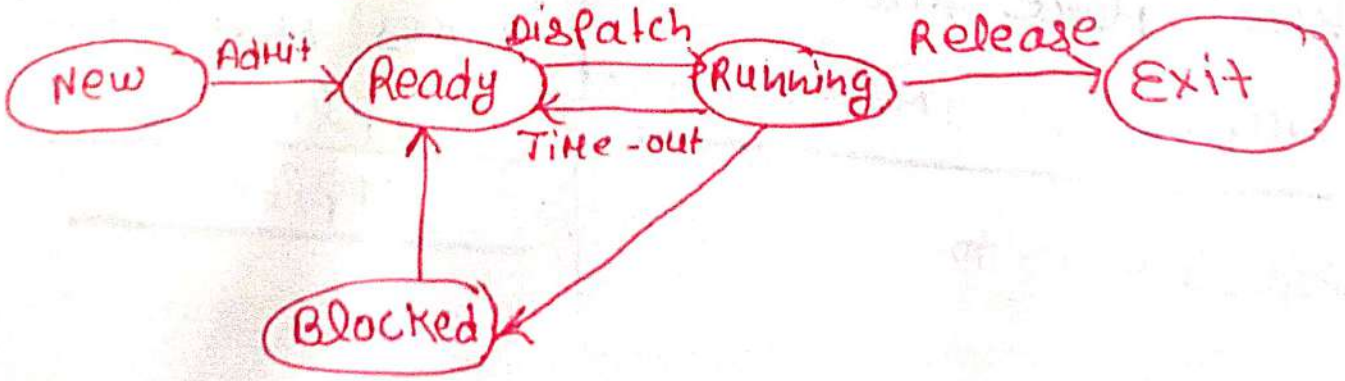
• The Process state is an indicator of the nature of the current activity in a Process.

• the notion of Process state is introduced to simplify Control of Process by the OS.



New :-

Or



1) New / Created :-

→ When a process is first created, it occupies the "Created" or "new" state.

→ In this states, the process awaits admission to the "ready" state.

→ Admission will be approved or delayed by a long-term scheduler (LTS) or job scheduler.

New → LTS → Ready

2) Ready :-

→ A "ready" or "waiting" process has been loaded into main memory and is awaiting execution on CPU.

→ In ready state, we will have multiple no. of process and one of the processes well be selected for scheduling on ~~"the"~~ the "running" state.

→ Short term scheduler (STS) or CPU scheduler selects one of the processes from ready state and allocates CPU for execution.

Ready → STS → Running

3) Running :-

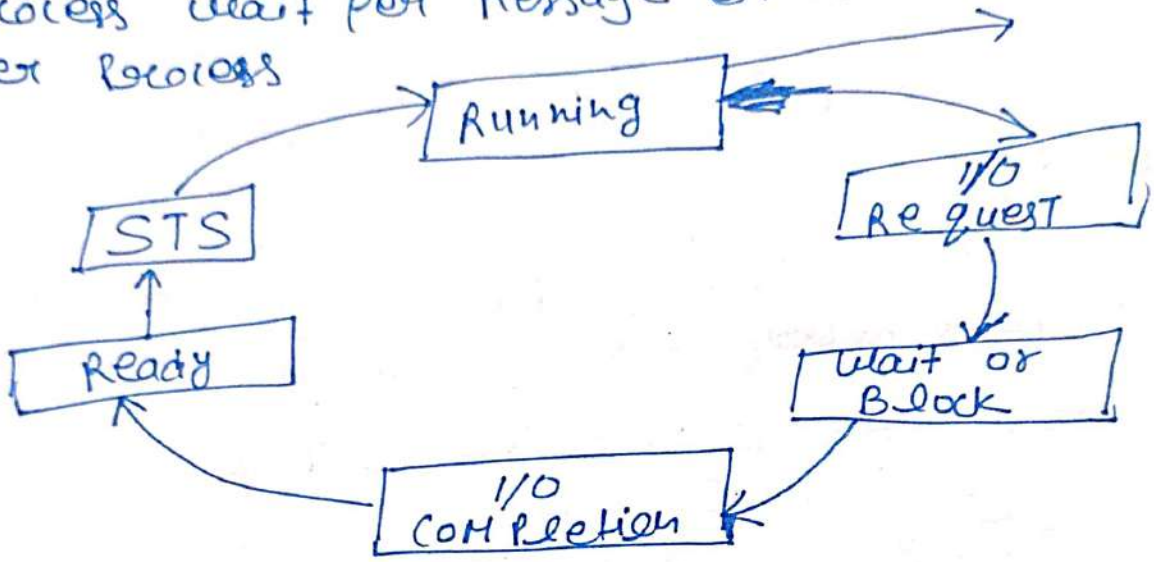
→ A process moves into the running state when chosen for execution.

→ The process is preempted because the OS decides to schedule some other higher priority process. → Ready

4) Wait or Block :-

→ the running process, sometimes move to waiting or block state because of following.

- → process require I/O operation.
- → process request memory and some other resources.
- → process wishes to wait for a specific interval of time.
- → process wait for message or action from some other process



5) Termination :-

→ A process may be terminated. either from the "running" state by completing its execution or by explicitly being killed.

Running → termination

ADDITIONAL Process States

1) Suspend Ready :-

→ When the resources are not sufficient to run the processes in ready state then some of the process will be suspended and move to 'Suspend Ready' state.

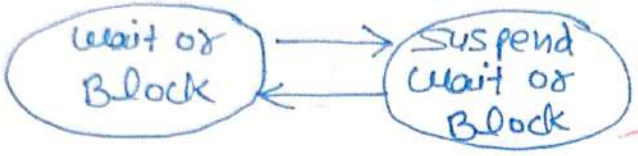
→ A process may be swapped out, that is, removed from main memory and placed on external storage by the mid term scheduler (MTS).

[Main (M)] → [MTS] → [Secondary (M)]

2) **Suspend and Block :-** → Also Called Swapped out and Blocked.

→ In this event the Process is both Swapped out and blocked.

→ Although in this case, the Process will Move to the blocked State, and May still be waiting for a resource to become available.

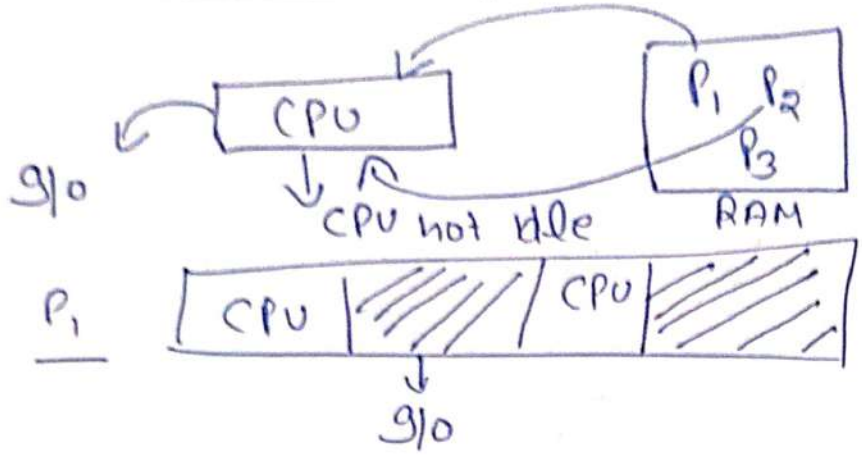


Processes :- A Proces is a Program in Execution including the Current values of variables registers, & Program Counter.

Program - group of Instructions

Process - Activity

Multiprogramming (OS) :- (non-preemptive)



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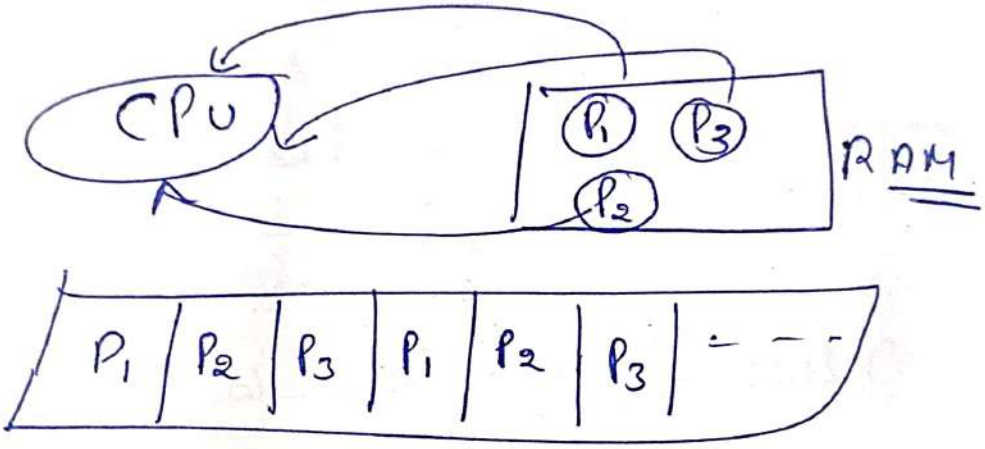
1) Responsive time is max for the last process.
2) Starvation. (waiting)

Multitasking / Time sharing OS :- (work with Time Quantum)



⇒ Embedded System :-

Specialized OS designed to perform a specific task for a device that is not a computer.

eg → Android OS
→ IOS. (iphone operating system)

Application → Smart TV, Smart washing Machine,
~~Smart A~~ (2) Traffic Lights.
(3) ATM Machines.
(4) Digital Cameras
(5) Air traffic Control.

Process state diagram

A process state diagram visually represents the different states a process can be in during its lifecycle and the transitions between them. Common states include New, Ready, Running, Blocked/Waiting, and Terminated. The diagram helps understand how processes are managed by an operating system, including scheduling and memory allocation.

Process State Diagram Notes:

- **New:**

The initial state of a process, where it's just created and its process control block (PCB) is created, but it's not yet in memory.

- **Ready:**

A process in the Ready state is waiting to be assigned to a CPU and is ready to execute.

- **Running:**

The process is currently being executed by the CPU.

- **Blocked/Waiting:**

The process is waiting for an event to occur, such as the completion of an I/O operation, a signal, or a resource.

- **Terminated:**

The process has completed its execution or has been terminated by the system.

- **Suspended (Ready/Waiting):**

A process may be moved to a suspended state (Ready Suspended or Waiting Suspended) when the system needs to reduce memory usage by swapping out a process's memory pages to secondary storage.

Process State Transitions:

- **New to Ready:**

When a new process is created, it transitions from the New state to the Ready state, waiting for CPU time.

- **Ready to Running:**

The operating system's scheduler selects a process from the Ready queue and assigns it to the CPU, transitioning it to the Running state.

- **Running to Ready:**

If the process's time slice is exhausted or a higher-priority process becomes ready, the running process is moved back to the Ready state.

- **Running to Blocked/Waiting:**

If the running process needs to wait for an event (e.g., I/O completion), it transitions to the Blocked/Waiting state.

- **Blocked/Waiting to Ready:**

Process state diagram

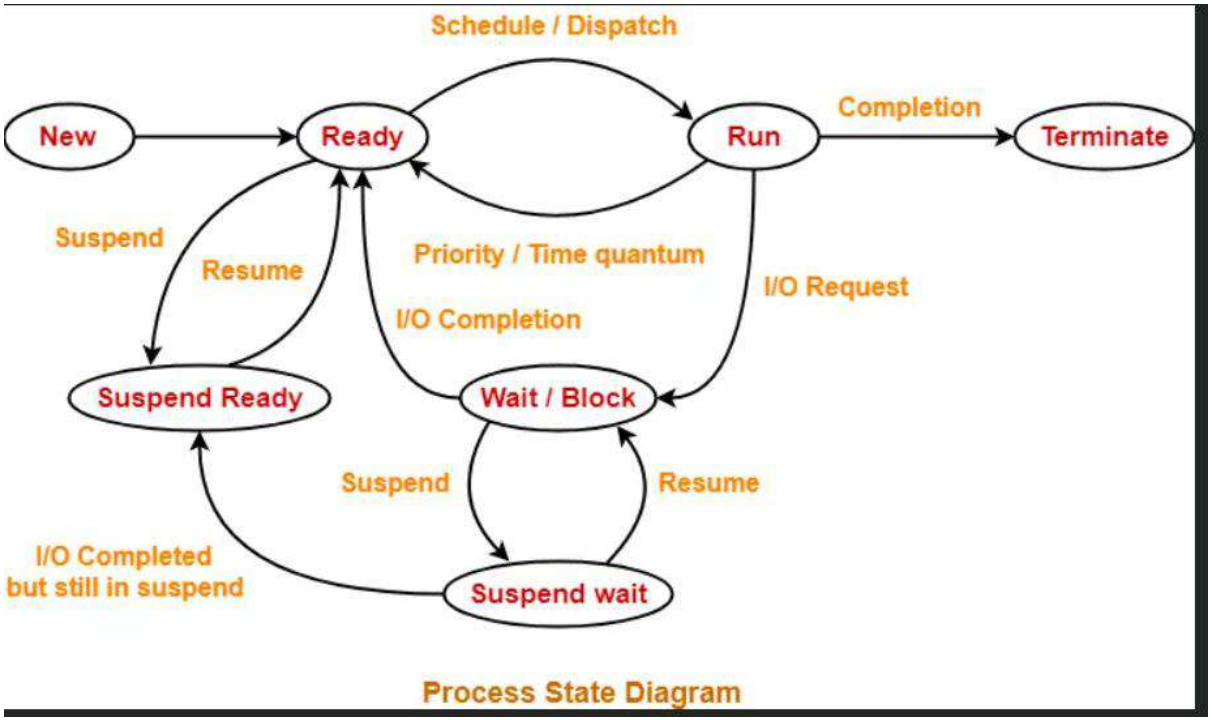
Once the event the process was waiting for occurs, it transitions back to the Ready state.

- **Running to Terminated:**

The process completes its execution and moves to the Terminated state.

Example:

1. A new process (New) is created and loaded into memory.
2. It moves to the Ready state, waiting for CPU time.
3. The scheduler selects the process, and it transitions to the Running state.
4. The process executes for a while.
5. It might then need to wait for an I/O operation, moving to the Blocked/Waiting state.
6. Once the I/O operation completes, it returns to the Ready state and can be scheduled again.
7. Finally, the process finishes and transitions to the Terminated state.



CPU Scheduling Algorithm :-

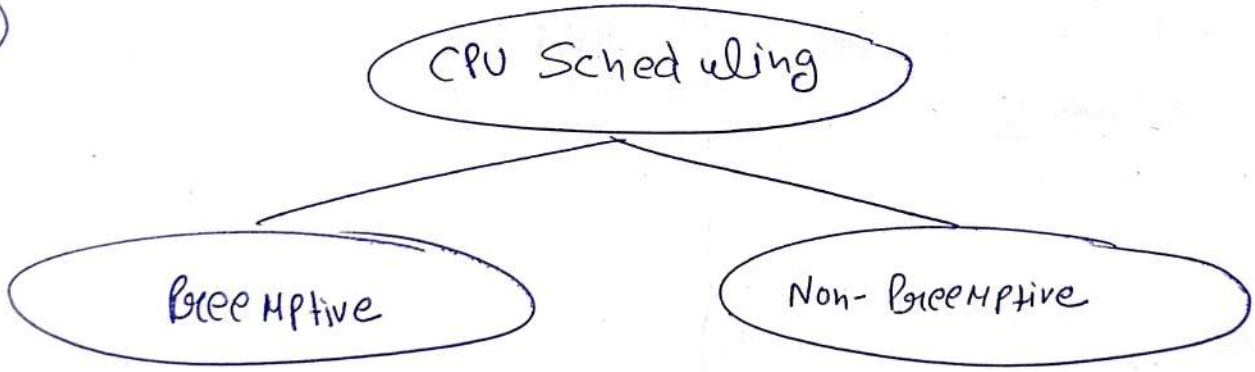
Scheduling

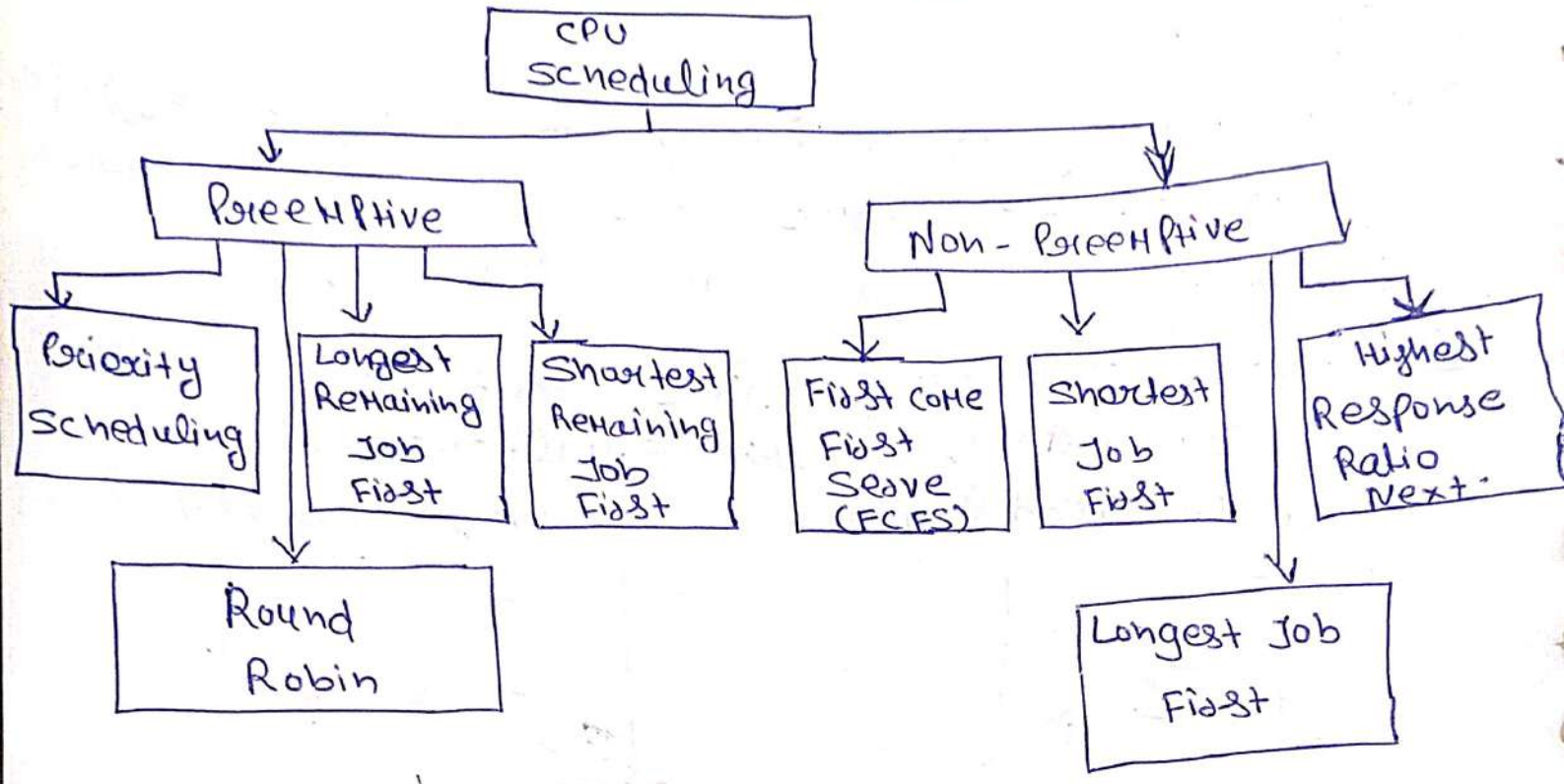
A Method of assigning CPU to a Process.

1) CPU scheduling is a process which allows one process to use the CPU while the execution of another process is on hold (in waiting state) due to unavailability of any resource like (I/O) etc.

2) The aim of CPU scheduling is to make the system efficient, fast and fair.

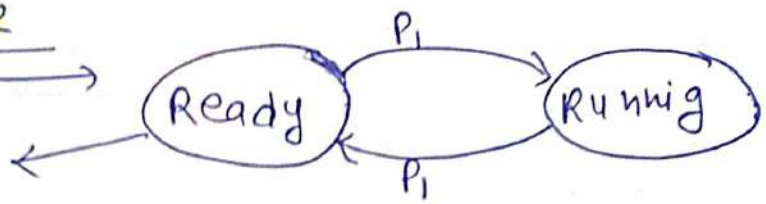
3)





Preemptive Scheduling	Non-Preemptive Scheduling
1) In this resources (CPU cycle) are allocated to a process for a limited Time.	1) once resources (CPU cycle) are allocated to a process, the process holds it till it completes its burst time or switches to waiting state.
2) if a process having high priority frequently arrives in the ready queue, a low priority process may starve (starvation).	2) if a process with a long burst time is running CPU, then later coming process with less CPU Time may starve.
3) Preemptiv Scheduling Responce Time is Less.	(3) ..... Responce (T) is high.
4) ~~The OS has~~ eg	4) eg

Pre-Emptive



→ Base responce (T)

→ Priority Base.

Terminologies used :-

1) Arrival Time :- The Time at which Process enter the Ready Queue or State.

2) Burst Time :- The Time required by a Process to get execute CPU.

3) Completion Time :- The Time at which Process Complete its execution.

4) Turnaround Time :- Completion (T) - Arrival (T).

5) Waiting Time :- Turnaround (T) - Burst (time).

6) Response Time :-

The Time at which a Process get CPU first (T) - Arrival (T).

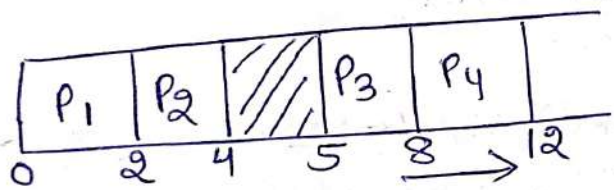
* Non Preemptive :-

1) FCFS :- It stand for first come first serve algo.

Process No	Arrival (T)	Burst Time or Executing (T)	Completion (T)	TAT	WT	RT
P_1	0	2	2	2	0	0
P_2	1	2	4	3	1	1
P_3	5	3	8	3	0	0
P_4	6	4	12	6	2	2
				$\frac{14}{4}$ ⇓ 3.5	$\frac{3}{4}$ ⇓ 0.75	$\frac{3}{4}$ ⇓

Criteria → "Arrival Time"

Mode → Non-Preemptive



Gantt Chart

TAT :- Turn around (T) is that per how much Time a Process stayed in the System.

Completion (T) - Arrival (T)

WT :- WT is ... how much was the useful Time and How much Time it had spent there.

$$\boxed{\text{TAT (T)} - \text{Burst (T)}}$$

RT :- Response Time = Time at which the Process got the CPU First (T) - Arrival (T).